

Information Security

Module 01

Duration: 26:44

Video: <https://www.youtube.com/watch?v=q4BLwk2kOGs>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Dr Navdeep Singh, Department of Computer Engineering, Punjabi University Patiala

Auto-generated YouTube transcript for personal study. Spoken wording belongs to the original lecture. Follow the video link above; this file is not an official CEC/SWAYAM publication.

0:00 [Music]

0:11 my dear students hello I welcome you all in our series of lectures on cyber security today we are going to discuss

0:20 about social and ethical issues of information security so let's start with it ethics

0:28 is a field of study that is concerned with distinguishing right from wrong and good from bad it analyzes the morality

0:35 of human behaviors policies laws and social structures ethicists attempt to justify

0:43 their moral judgments by reference to ethical principles of theories that attempt to capture our moral intuitions

0:51 about what is right and wrong the two theoretical approaches that are most common in ethics are

0:58 consequentialism and analogy consequentialist approaches assume that actions are wrong to the

1:06 extent that they have bad consequences whereas dlal approaches assume that people have moral duties that exist

1:15 independently of any good or bad consequences that their actions may have ethical principles often inform

1:24 legislation but it is recognized in ethics that legislation cannot function as a substitute for

1:32 Morality it is for this reason that individuals and corporations are always required to consider not only the

1:39 legality but also the morality of their actions ethical analysis of security and

1:48 privacy issues in Information Technology primarily takes place in computer ethics

1:54 which emerged in the 1980s as a field computer eth ethics analyzes moral responsibilities of computer

2:03 professionals and computer users and ethical issues in public policy for information technology development and

2:11 use it asks such questions as is it wrong for corporations to read their employees email or is it morally

2:20 permissible for computer users to copy copyrighted software or should people be free to put controversial or

2:27 pornographic content online with without censorship ethical issues and questions like these require moral or ethical

2:36 analysis analysis in which the moral dilemmas contained in these issues are clarified and solutions are proposed for

2:45 them moral analysis aims to get clear on the facts and values in such cases and

2:52 to find a balance between the various values rights and interests that are at stake and to propose or evaluate

3:00 policies and courses of action now the moral importance of computer

3:07 security computer security is a field of computer science concerned with the application of security features to

3:14 computer systems to provide protection against the unauthorized disclosure manipulation or deletion of information

3:22 and against denial of service the condition resulting from these efforts is also called computer security the aim

3:29 of computer Security Professionals is to attain protection of valuable information and system

3:37 resources a distinction can be made between the security of the system resources and the security of

3:44 information or data the first may be called system security and the second information security or data

3:52 security system security is the protection of the hardware and software of a computer system against malicious

4:00 programs that sabotage system resources information security is the

4:07 protection of data that resides on disk drives on computer systems or is transmitted between

4:14 systems information security is customarily defined as concerned with the protection of three aspects of data

4:23 confidentiality integrity and availability now how does computer security pose ethical

4:31 issues as explained earlier ethics is mostly concerned with rights harms and interest we may therefore answer this

4:40 question by exploring the relation between computer security and rights harms and

4:46 interest now the important questions are what morally important benefits can computer security bring what morally

4:54 important harms or violations of moral rights can result from a lack of computer security or can computer

5:02 security also cause harms or violate rights instead of preventing and protecting them a first and perhaps most obvious

5:11 harm that can occur from breaches of computer security is economic harm when system security is undermined

5:19 valuable hardware and software may be damaged or corrupted and service may become unavailable resulting in losses of time

5:28 money and sources breaches of information security may come at an even higher economic cost valuable data may

5:36 be lost or corrupted that is worth much more than the hardware on which it is stored and this may cause severe economic

5:43 losses stored data may also have personal cultural or social value as opposed to the economic value that can

5:51 be lost when data is corrupted or lost any type of loss of system or data security is more likely to cause some

5:59 amount of psychological or emotional harm breaches of computer security may even cause grave harms like injury and

6:09 death this may occur in so-called safety critical systems which are computer systems with a component or realtime

6:17 control that can have a direct life-threatening impact examples are computer systems in

6:24 nuclear reactor control aircraft and air traffic control missile systems and medical treatment systems the

6:32 corruption of certain other types of systems may also have life-threatening consequences in a more indirect way

6:39 these may include systems that are used for design monitoring diagnosis or decision making for instance systems

6:47 used for bridge design or medical diagnosis compromises of the confidentiality of information may cause

6:55 additional harms and rights violations third parties May compromise the confidentiality of information by

7:02 accessing copying and disseminating it such actions May first of all violate property rights including intellectual

7:10 property rights which are rights to own and use intellectual Creations such as artistic or literary works and

7:17 Industrial designs the information may be exclusively owned by someone who has the right to determine who can access and

7:25 use the information and this right can be violated second compromises of confidentiality

7:34 May violate privacy rights this occurs when information that is accessed includes information about persons that

7:42 is considered to be private in addition to violations of property and privacy rights breaches of

7:49 confidentiality may also cause a variety of other harms resulting from the dissemination and use of confidential

7:57 information for instance dissemination of internal members of a firm damages its reputation and compromises of the

8:05 confidentiality of online credit card transactions undermines trust in the security of online Financial

8:12 transactions and harms ebanking and Ecommerce activity compromises of the availability

8:19 of information can when they are prolonged or intentional violate Freedom rights specifically rights to Freedom of

8:27 Information and Free Speech Freedom of Information is the right to access and use public

8:34 information shutting down Vital Information Services could violate this right to information in addition computer

8:43 networks have become important as a medium for speech websites email bulletin boards

8:50 and other services are widely used to spread messages and communicate with others when access to such Services is

8:58 blocked for instance through denial of service attacks or hijackings of websites such acts are properly

9:05 classified as violations of free speech computer security measures normally prevent harms and protect rights but

9:13 they can also cause harm and violate rights notably security measures may be

9:20 so protective of information and system resources that they discourage or prevent stakeholders from accessing

9:27 information or using services security measures may also be discriminatory they may wrongly exclude

9:34 certain classes of users from using a system or may wrongly privilege certain classes of users over

9:42 others next is ethical issues in computer security number one hacking and computer

9:50 crime a large part of computer security is concerned with the protection of Computer Resources and data against

9:58 unauthorized intention breakings or disruptions such action is often called hacking hacking is the use of computer

10:06 skills to gain unauthorized access to Computer Resources hackers are highly skilled

10:13 computer users that use their talents to gain such access and often from communities or networks with other

10:20 hackers to share knowledge and data hacking is often also defined more negatively
as the gaining of such

10:29 unauthorized access for Malaysia purposes to steal information and software or to
corrupt data or disrupt

10:37 system operations self-identified hackers however make a distinction between
non-malicious break-ins which

10:44 they describe as hacking and malicious and disruptive break-ins which they call
cracking self-identified hackers often

10:52 justify their hacking activities by arguing that they cause no real harm and instead
have a positive EMP

11:00 fact the positive impact of hacking they argue is that it frees data to the benefit of
all and improve systems and

11:08 software by exposing security holes these considerations are part of what has been
called The Hacker ethic or

11:17 hacker code of ethics which is a set of principles that guide the activity of many

11:24 hackers such principles include convictions that information should be free that
access to computers should be

11:31 unlimited and total and that activities in cyberspace cannot do harm in the real

11:38 world the belief that information should be free runs counter to the very notion of
intellectual property and would imply

11:47 that creators of information would have no right to keep it to themselves and have
no opportunity to make a profit from

11:55 it it would more however fundamentally undermine priv privacy and would
undermine the integrity and accuracy of

12:03 information as information could be modified and changed at will by anyone who
would access

12:10 it there are many varieties of computer crime and not all of them compromise
computer security there are two major types of

12:18 cyber crime that compromise computer security cyber trespass which is defined as
the use of Information Technology to

12:25 gain unauthorized access to computer systems or password protected websites and
cyber vandalism which is the use of

12:33 Information Technology to unleash programs that disrupt the operations of
computer networks or corrupt

12:41 data number two cyber terrorism and information Warfare a recent concern in
computer and

12:49 National Security has been the possibility of cyber terrorism which is defined as the
execution of politically

12:56 motivated hacking operations intended to cause grave harm that is resulting in
either loss of life or severe economic

13:04 loss or both the possibility of major attacks on information infrastructure intending to

13:11 debilitate or compromise this infrastructure and harm economic industrial or social structures

13:17 dependent on it has become a major concern since the 9/11 tax such attacks could be both foreign

13:26 and domestic cont y exist on the proper scope of cyber terrorism where should the boundaries be

13:34 drawn between cyber terrorism cyber crime and cyber vandalism should a teenager who releases

13:42 a dangerous virus that turns out to cause major harm to government computers be persecuted as a cyber

13:50 terrorist or are politically motivated hijackings of the homepages of major organizations acts of cyber terrorism

13:59 a distinction between cyber terrorism and other kinds of cyber attacks may be found in its political nature cyber

14:06 terrorism consists of politically motivated operations that aim to cause harm there is a distinction between

14:13 cyber terrorism and hactivism activism are hacking operations against an Internet site or server with the

14:21 intent to disrupt normal operations but without the intent to cause serious damage

14:29 activists may make use of email bombs low grade viruses and temporary homepage hijackings there are politically motivated hackers who engage in a form of

14:36 electronic political activism that should be distinguished from

14:45 terrorism information Warfare is an extension of ordinary Warfare in which combatants use information and attacks

14:53 on information and information systems as tools of warfare information Warfare May include the use of information media

15:01 15 minutes, 1 second to spread propaganda the disruption jamming or hijacking of communication infrastructure or propaganda fears of

15:08 the enemy and hacking into computer systems that control vital infrastructure for example oil and gas

15:15 pipelines electric power grids or rail infrastructure now what are the moral responsibilities of Information Security

15:25 Professionals Information Security Professionals are individuals whose job is to maintain system and information

15:32 security by standing of their profession they have a professional responsibility to assure the correctness reliability

15:40 availability Safety and Security of all aspects of information and information

15:47 systems the discussion makes clear that this responsibility has a moral Dimension
professional activities in

15:55 computer security may protect people from AI important harms but could also
cause such harms and may either protect

16:04 or violate people's moral rights in case of safety critical systems the decisions of
important

16:11 Security Professionals may even be a matter of life or death that information
Security Professionals have moral

16:18 responsibilities as part of their profession is reflected in CES of Ethics used by
various organizations for

16:25 computer and information security these CES of Ethics rarely go into detail
however on the moral

16:34 responsibilities of Information Security Professionals in specific situations for
instance the code of

16:41 ethics of Information Systems security Association an international organization of
Information Security

16:48 Professionals and practitioners only states that members should perform all
professional activities and duties in

16:56 accordance with all applicable laws and the highest ethical principles but does not
go on to specify what these ethical

17:04 principles are or how they should be applied and balanced against each other in
specific

17:12 situations for Information Security Professionals as well as for other computer
professionals who have a responsibility for computer security a

17:20 code of ethics clearly is not enough to appreciate the moral dimension of their work
and to cope with moral dilemmas in it

17:30 they require training in information security ethics such training help professionals
to get clear about interests rights and

17:39 moral values that are at stake in computer security to recognize ethical questions
and dilemmas in their work and

17:47 to balance different moral principles in resolving such ethical issues number three
information privacy

17:56 and ethics what is privacy and why is it important in Western societies a broad
recognition exist of a right to

18:04 personal privacy the right to privacy was first defended by the American Justice
Samuel Brandeis who

18:12 defined privacy as the right to be let alone privacy is a notion that is difficult to
Define and many more

18:20 precise definitions have since been presented often the right to privacy is defined
as a right of individuals to

18:28 Control Act access or interference by others into their Private Affairs privacy is held to be valuable

18:35 for several reasons most often it is held to be important because it is believed to protect individuals from all kinds of

18:43 external threats such as defamation ridicule harassment manipulation blackmail theft subordination and

18:52 exclusion it has also been argued that privacy is a necessary condition for autonomy

18:59 without privacy people could not experiment in life and develop their own personality and own thoughts because they would constantly be subjected to

19:08 the judgment of others the right to privacy has also been claimed to protect other rights such as abortion rights and the right to

19:17 sexual expression privacy moreover has been claimed to have social value in addition

19:23 to individual value it has for instance been held to be essential for maintaining democracy the right to privacy is not

19:32 normally held to be absolute it must be balanced against other rights and interest such as the maintenance of

19:39 public order and national security privacy rights may also vary in different contexts there is for example

19:48 a lesser expectation of privacy in the workplace or in the public sphere than there is at

19:55 home an important principle used in privacy protection in Western Nations is that of informed consent it

20:02 is often held that citizens should be informed about how organizations plan to store use or exchange their personal

20:10 data and that they should be asked for their consent people can then voluntarily give up their privacy if they

20:19 choose number four information technology and privacy privacy is a value in modern

20:27 societies that corresponds with the ideal of the autonomous individual who is free to act and decide his own

20:34 destiny yet modern societies are also characterized by surveillance a practice that tends to undermine

20:42 privacy surveillance is a systematic observation of groups of people for specific purposes usually with the aim

20:50 of exerting some form of influence over them the state engages in surveillance to protect National Security and to

20:58 fight crime and the modern Corporation engages in surveillance in the workplace to retain control over the

21:06 workforce computerization from the 1960s onward has intensified surveillance by increasing its scale ease and

21:15 speed surveillance is partially delegated to computers that help in collecting processing and exchanging

21:22 data computers have not only changed the scale and speed of surveillance they have also made a new kind of

21:29 surveillance possible data wance which is the large scale computerized collection and processing of personal

21:36 data in order to monitor people's action and Communications more and more information

21:43 technology is not just used to record and process static information about individuals but to record and process

21:50 their actions and communication new detection Technologies like smart Clos circuit televisions Biometrics and intelligent user interfaces

21:59 and new data processing techniques like data mining further exaggerate this trend the ease with which surveillance

22:07 now takes place has made it a generalized activity that is routinely performed in all kinds of settings by

22:14 different kinds of organizations corporations for instance have extended surveillance from their workplace to their customers called

22:22 consumer surveillance in addition the 911 terrorist attacks have drastically Ally expanded surveillance activities by the

22:32 state many privacy disputes in today's society result from tensions between people's right to privacy and state and

22:39 corporate interest in surveillance in the Information Society privacy protection is realized through all kinds of information privacy laws

22:48 policies and directives or data protection policies these policies regulate the harvesting processing

22:56 usage storage and exchange of personal data they're often overtaken however by new developments in

23:05 technology however privacy protection has also become a concern in the design and development of Information

23:13 Technology number five privacy in public it is sometimes believed that privacy is a right that people have when

23:22 they are in private places like homes private clubs and restrooms but that is minimized or for fed as soon as they

23:29 enter public space when you walk in public streets or are on the road with your car it is sometimes believed you

23:37 may retain the right not to be seized and searched without probable cost but your appearance and behavior may be

23:44 freely observed surveilled and registered many privacy Scholars however have argued that this position is not

23:52 fully tenable and that people have privacy rights in public areas that are incompatible with with certain registration and surveillance

24:01 24 minutes, 1 second practices the problem of privacy in public applies to the tracking recording and surveillance of public appearances

24:09 movements and behaviors by individuals and their vehicles techniques that are used for this including video

24:16 surveillance including smart CCTV for facial recognition infrared cameras satellite

24:24 surveillance GPS tracking are F ID tagging electronic

24:32 checkpoints mobile phone tracking audio bugging and mate intelligence

24:40 techniques number six the last but not the least is biometric identification Biometrics is the

24:48 identification or verification of someone's else identity on the basis of psychological or behavioral

24:55 characteristics biometric Technologies provide a reliable method of access control and personal identification for

25:03 governments and organizations however Biometrics has also raised privacy concern widespread

25:11 use of Biometrics would have the undesirable effect of eliminating anonymity and pseudonymity in most daily

25:18 transactions because people would leave unique traces everywhere they go more however the biometric monitoring of

25:25 movements and actions gives the monitoring organization ation insight into a person's behaviors which may be used against that person's

25:34 interests in addition many people find biometrics distasteful because it involves the recording of unique and

25:42 intimate aspects of a person and because biometric identification procedures are sometimes invasive of bodily

25:50 privacy the challenge for Biometrics is therefore to develop techniques and policies that optimally protective of

25:57 personal PR pracy with this our today's lecture is over concluding the lecture we have learned about what are ethics

26:05 importance of computer security and various social and ethical issues hope today's lecture is clear to you in the next lecture we shall learn about

26:13 Information Security Management till then goodbye have a nice day

26:28 n

26:32 [Music]